## Profile

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**GitHub** | **Notion**

## Education

**2017.03 – 2025.05**
Daejeon University | Biology | degree incomplete
**2026.03 – 2026.10**
Daejeon DW Academy AIoT-based Drone Video Monitoring System Development
(Intensive Software Engineering Bootcamp)

# Portfolio Index

# Data Analysis

## Analysis of How Genre and Release Timing Affect Steam Game Success (Positive Review Rate & User Base) (Personal Project)

**GitHub:** https://github.com/yehyun-42/yehyun-42.github.io/tree/main/Project/SQL_project

### Tech Stack

| | |
|---|---|
| Language | Python, SQL |
| IDE | Jupyter, MySQL |
| Data & Viz | Pandas, Matplotlib |

**Dataset**: **Steam Market and Product Metadata**
(Source: Kaggle)

**Observation**: Instances have been identified where games perceived as having high quality failed to achieve commercial success.
**Issue**: Since titles from new developers sometimes succeed, the developer's brand recognition alone cannot explain the outcome. It is necessary to identify factors other than the developer that influence commercial success, as well as to understand the direct relationship between the developer and a game's market performance.
**Expected Outcome**: By identifying the conditions necessary to mitigate risk for new game titles, it is anticipated that developers can generate gamer interest and achieve a certain level of revenue.

#### Reasons for Selecting the Dataset

In the case of Korean games, the number of available titles is limited, and information is fragmented because each game company operates its own platform. Consequently, it was decided to select a dataset from Steam—an integrated distribution platform—via Kaggle.

#### Hypothesis

- Commercial success will vary depending on the game genre.
- There will be a proportional relationship between the number of users and average ratings.
- Games developed by well-known companies will have a larger user base.
- Lower prices will increase accessibility, resulting in a larger user base.
- Games released longer ago will have higher exposure, leading to a larger user base.

# Data Analysis Results

**Game success by genre**

- Distinct differences in estimated owners and average ratings were observed depending on the game genre.
- The **action genre** recorded the highest figures for user numbers, while the **indie genre** showed the highest average ratings.

**Proportional relationship between the estimated owners and average ratings**

- **Indie games** have the highest average ratings, while **free-to-play games** have the highest average player counts.
- However, free-to-play games have the lowest average ratings.
- Rather than indicating that these games failed to meet expectations, this likely reflects their high accessibility; **a wide range of users often tries them out casually, regardless of their personal preferences.**

**estimated owners according to developer awareness**

- **Developers with high brand** recognition tend to **have a large average user base**.
- However, the fact that average ratings sometimes fall below 75% suggests that a significant number of purchases are **driven solely by the developer's name.**
- Nevertheless, since these titles generally maintain a "Positive" rating on Steam, it can be concluded that purchases based solely on the **developer's name still result in relatively high levels of satisfaction.**

**estimated owners according to price**

- Games with both large and small player bases **exhibit a diverse range of price** points.
- It appears **that price does not significantly influence a game's success.**

**estimated owners according to release date**

- Sorting the games by user count in ascending order and examining their release dates revealed that the list includes not only recently released titles but also games launched over a decade ago.
- While the release date appears to have some influence, **it does not seem to be a strictly proportional factor.**

# Conclusion

An analysis of user density and rating distribution on the Steam platform yielded insights for selecting target genres and setting price points for new game launches.

The analysis suggests that a genre's mass appeal has a greater impact on performance than factors such as the release date or price. Additionally, a developer's brand recognition serves to raise the baseline level of trust in a new game.

For new developers launching a new title, prioritizing genre popularity over price or release timing offers a more competitive advantage.

However, considering examples like Krafton's 'inZOI', aiming to position a game as a viable alternative within a genre that currently lacks substitutes is also considered a valid strategy.

# Data Preprocessing

### Data Used for Hypothesis Testing

- app_id and name. release_data. price_usd. review_score_pct. total_reviews. estimated_owners. steam_official_genres. steam_developer.

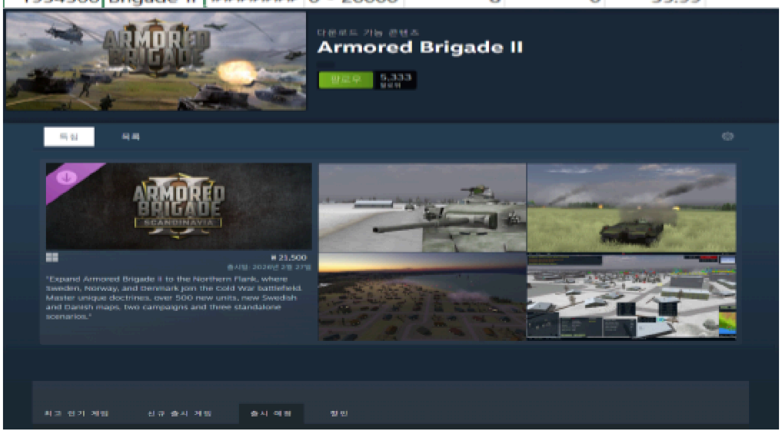
### Data Preprocessing & Analysis Standard

- Analysis Standard
  Positive review(review_score_pct) criteria: Steam's "Overwhelmingly Positive" standard.
  (an **average of 95% or more positive reviews**.)
- Outliers Processing Policy
  Processing: **Do not remove**
  Purpose of setting Processing Policy: The possibility that the games actually received overwhelmingly positive or negative ratings cannot be ruled out.
  Accordingly, the decision was made not to remove outliers.

### First try

- After completing preprocessing steps, exported the data to a CSV file using 'to_csv', but the DataFrame columns became misaligned.
- Attempted the preprocessing again using 'read_excel' instead of 'read_csv', but discovered a flaw within the DataFrame itself.
- To verify this, checked the DLC for 'Armored Brigade II' which showed a count of 10 in the "discount DLC" column, and confirmed that only one DLC has actually been released.
- It was determined that the CSV DataFrame itself is unsuitable for data analysis due to reliability issues.
- Decided to use a reliable dataframe instead.

| AppID | Name | Release dá | Estimated | Peak CCU | Required á | Price | DiscountD | About the | Supported | Full audio | Reviews |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2539430 | Black Drag | ####### | 0 - 0 | 0 | 0 | 0 | 0 | 0 | | [] | [] |
| 496350 | Supipara - | 29-Jul-16 | 0 - 20000 | 0 | 0 | 5.24 | 65 | 0 | Springtime | ['English'] | [] |
| 1034400 | Mystery S | ####### | 0 - 20000 | 0 | 0 | 4.99 | 0 | 0 | Immerse y | ['English', ' | [] |
| 3292190 | 踰꾨쾭踰?: | ####### | 0 - 20000 | 1 | 0 | 8.99 | 0 | 1 | synopsis 'ㅏ | ['Korean'] | ['Korean'] |
| 3631080 | Maze Que | ####### | 0 - 20000 | 0 | 0 | 4.99 | 0 | 0 | Its not just | ['English'] | ['English'] |
| 1654170 | Agony VR | ####### | 0 - 20000 | 0 | 0 | 13.99 | 0 | 0 | A JOURNE | ['English', ' | ['English', '' |
| 1934300 | Brigade II | ####### | 0 - 20000 | 8 | 0 | 35.99 | 10 | 1 | Building o | ['English'] | [] |




Discovered flaws in the DataFrame itself after completing preprocessing

**Second try**

- After completing preprocessing steps the data was exported as a table using 'to_csv'.
- Separated into a user table and a genre table for analysis.
- For the genre table, if a single row contains multiple genres, there is a risk that the same genre could be classified as distinct; therefore, each genre is separated into its own row.

```python
df_final['name'] = df_final['name'].astype(str).str.replace(r"[^a-zA-Z0-9\s]", "", regex=True)
df_final['steam_developer'] = df_final['steam_developer'].astype(str).str.replace(r"[^a-zA-Z0-9\s]", "", regex=True)
df_final.head(20)
```

| | app_id | name | release_date | price_usd | review_score_pct | total_reviews | estimated_owners | steam_developer | review_score_negative |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 730 | CounterStrike 2 | 2012-08-21 | 0.00 | 83 | 4980365 | 149410950 | Valve | 17 |
| **1** | 2868840 | Slay the Spire 2 | 2026-03-05 | 24.99 | 97 | 49549 | 1486470 | Mega Crit | 3 |
| **3** | 3065800 | Marathon | 2026-03-05 | 39.99 | 90 | 24360 | 730800 | Bungie | 10 |
| **4** | 3764200 | Resident Evil Requiem | 2026-02-26 | 69.99 | 96 | 79667 | 2390010 | CAPCOM Co Ltd | 4 |
| **5** | 2157830 | John Carpenters Toxic Commando | 2026-03-12 | 39.99 | 83 | 2765 | 82950 | Saber Interactive | 17 |
| **6** | 1144200 | Ready or Not | 2023-12-13 | 49.99 | 85 | 243810 | 7314300 | VOID Interactive | 15 |
| **7** | 1172470 | Apex Legends | 2020-11-04 | 0.00 | 65 | 1577 | 47310 | Respawn | 35 |
| **8** | 1808500 | ARC Raiders | 2025-10-30 | 39.99 | 79 | 264150 | 7924500 | Embark Studios | 21 |
| **9** | 2852190 | Monster Hunter Stories 3 Twisted Reflection | 2026-03-12 | 69.99 | 75 | 1570 | 47100 | CAPCOM Co Ltd | 25 |
| **10** | 1167630 | Teardown | 2022-04-21 | 14.99 | 94 | 115777 | 3473310 | Tuxedo Labs | 6 |
| **11** | 2552430 | KINGDOM HEARTS HD 1525 ReMIX | 2024-06-13 | 24.99 | 89 | 8167 | 245010 | Square Enix | 11 |
| **12** | 4069520 | Super Battle Golf | 2026-02-19 | 7.99 | 95 | 4025 | 120750 | Brimstone | 5 |
| **14** | 1222670 | The Sims 4 | 2020-06-18 | 0.00 | 79 | 82661 | 2479830 | Maxis | 21 |
| **15** | 2778610 | Over The Top WWI | 2026-03-06 | 16.26 | 88 | 3572 | 107160 | Flying Squirrel Entertainment | 12 |
| **16** | 359550 | Tom Clancys Rainbow Six Siege | 2015-12-01 | 0.00 | 75 | 1234092 | 37022760 | Ubisoft Montreal | 25 |
| **17** | 553850 | HELLDIVERS 2 | 2024-02-08 | 39.99 | 70 | 807861 | 24235830 | Arrowhead Game Studios | 30 |
| **18** | 230410 | Warframe | 2013-03-25 | 0.00 | 90 | 2880 | 86400 | Digital Extremes | 10 |
| **19** | 2357570 | Overwatch | 2023-08-10 | 0.00 | 60 | 257 | 7710 | Blizzard Entertainment Inc | 40 |
| **20** | 236390 | War Thunder | 2013-08-15 | 0.00 | 76 | 1961 | 58830 | Gaijin Entertainment | 24 |
| **21** | 3472040 | NBA 2K26 | 2025-09-04 | 19.99 | 74 | 12634 | 379020 | Visual Concepts | 26 |

```python
df_genres['name'] = df_genres['name'].astype(str).str.replace(r"[^a-zA-Z0-9\s]", "", regex=True)
df_genres['steam_official_genres'] = (
    df_genres['steam_official_genres']
    .astype(str)
    .str.replace(r"[\[\]'\"]", "", regex=True))
df_genres['steam_official_genres']=df_genres['steam_official_genres'].str.split(',')
df_genres=df_genres.explode('steam_official_genres')
df_genres['steam_official_genres']=df_genres['steam_official_genres'].astype(str).str.strip()

df_genres.head(20)
```

| | app_id | name | steam_official_genres |
|---|---|---|---|
| **0** | 730 | CounterStrike 2 | Action |
| **0** | 730 | CounterStrike 2 | Free To Play |
| **1** | 2868840 | Slay the Spire 2 | Indie |
| **1** | 2868840 | Slay the Spire 2 | Strategy |
| **1** | 2868840 | Slay the Spire 2 | Early Access |
| **2** | 3321460 | Crimson Desert | Action |
| **2** | 3321460 | Crimson Desert | Adventure |
| **3** | 3065800 | Marathon | Action |
| **4** | 3764200 | Resident Evil Requiem | Action |
| **4** | 3764200 | Resident Evil Requiem | Adventure |
| **5** | 2157830 | John Carpenters Toxic Commando | Action |
| **5** | 2157830 | John Carpenters Toxic Commando | Adventure |
| **6** | 1144200 | Ready or Not | Action |
| **6** | 1144200 | Ready or Not | Adventure |
| **6** | 1144200 | Ready or Not | Indie |
| **7** | 1172470 | Apex Legends | Action |
| **7** | 1172470 | Apex Legends | Adventure |
| **7** | 1172470 | Apex Legends | Free To Play |
| **8** | 1808500 | ARC Raiders | Action |
| **9** | 2852190 | Monster Hunter Stories 3 Twisted Reflection | Adventure |

Pre-processed table

# Hypothesis Testing

## Hypothesis 1. Commercial success will vary depending on the game genre

- Indie games have the highest average ratings, while free games have the lowest.
- In terms of estimated owner numbers by genre, the action genre has the highest count.

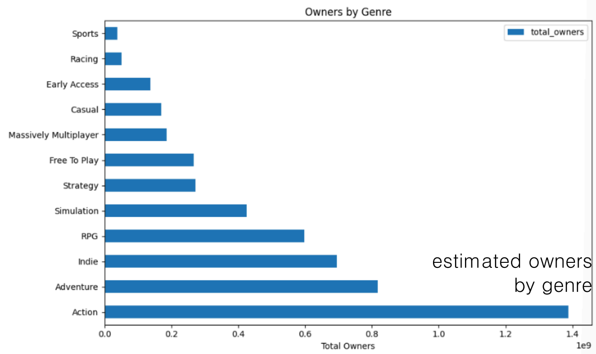
```sql
select n.steam_official_genres, avg(g.review_score_pct) as positive
from game_table g, genres_table n
where g.app_id=n.app_id
group by n.steam_official_genres
order by positive desc;
```

| steam_official_genres | positive |
|---|---|
| Indie | 87.1201 |
| Casual | 86.6970 |
| Simulation | 84.6833 |
| Adventure | 83.5282 |
| Strategy | 83.4787 |
| Early Access | 82.8696 |
| Action | 82.3141 |
| Racing | 81.9250 |
| RPG | 81.1877 |
| Sports | 78.3810 |
| Massively Multiplayer | 71.7969 |
| Free To Play | 69.5962 |

Sorted by average number of positive ratings per genre in descending order

```python
owners_by_genre.set_index('steam_official_genres')[['total_owners']].plot.barh(
    figsize=(10, 6), title='Owners by Genre', xlabel='Total Owners', ylabel=''
)

plt.tight_layout()
plt.show()
```




estimated owners by genre

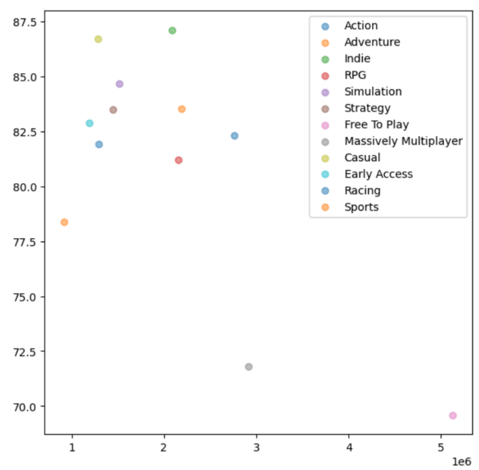
- Changes in the average value based on the estimated owner were considered; however, upon examining the lowest-rated free games and the sports genre—which has the fewest users— it was determined that the estimated owners had a negligible impact on the average.

## Hypothesis 2. There will be a proportional relationship between the number of users and average ratings.

- While free-to-play games boast the highest average estimated owners, indie games hold the highest average ratings.
- The high user count for free games appears to be driven more by price accessibility than by average ratings.

```python
fig=plt.figure(figsize=(7,7))
ax=fig.add_subplot(111)

for genres in genre_avg['steam_official_genres'].unique():
    sub=genre_avg[genre_avg['steam_official_genres']==genres]
    ax.scatter(x=sub['avg_owners'],
               y=sub['avg_positive'],
               alpha=0.5,
               label=genres)
ax.legend()
plt.show()
```




Scatter plot of estimated owners(x–axis) avg review score pct (y–axis) by genre

## Hypothesis 3. Games developed by well-known companies will have a larger user base.

- Renowned game companies (such as Bethesda, Fromsoftware, and CAPCOM) tend to have high average player counts.
- However, there is also a tendency for average ratings to sometimes fall below 75%.

- It has been determined that the name of the game developer alone is sufficient to secure a certain level of user volume.
- Even when average ratings are low, the game remains within the "Positive" range on Steam, indicating relatively high satisfaction among players who purchased it based solely on the developer's reputation.

```sql
select n.steam_official_genres,
    sum(g.estimated_owners) as total_owners,
    avg(g.estimated_owners) as avg_owners
from game_table g, genres_table n
where g.app_id=n.app_id
group by n.steam_official_genres
order by total_owners desc;
```

| steam_official_genres | total_owners | avg_owners |
|---|---|---|
| Action | 1388608890 | 2760653.8569 |
| Adventure | 817216260 | 2190928.3110 |
| Indie | 695782950 | 2089438.2883 |
| RPG | 597676890 | 2157678.3032 |
| Simulation | 425611110 | 1514630.2847 |
| Strategy | 272506170 | 1449500.9043 |
| Free To Play | 266590830 | 5126746.7308 |
| Massively Multiplayer | 186719040 | 2917485.0000 |
| Casual | 169345560 | 1282920.9091 |
| Early Access | 137200080 | 1193044.1739 |
| Racing | 51681870 | 1292046.7500 |
| Sports | 38629380 | 919747.1429 |

```sql
select steam_developer,
    avg(estimated_owners) as avg_owners,
    avg(review_score_pct) as avg_positive
from game_table
group by steam_developer
order by steam_developer asc;
```

| steam_developer | avg_owners | avg_positive |
|---|---|---|
| 10 Chambers | 1345200.0000 | 76.0000 |
| 11 bit studios | 582450.0000 | 70.0000 |
| 24 Entertainment | 5926740.0000 | 65.0000 |
| 343 Industries | 1022280.0000 | 58.0000 |
| 343 Industries Splash Damag... | 6985260.0000 | 85.0000 |
| Acute Owl Studio | 67020.0000 | 96.0000 |
| AdHoc Studio | 4716480.0000 | 95.0000 |
| Aesir Interactive | 534060.0000 | 73.0000 |
| Aesir Interactive NightinGames | 25470.0000 | 80.0000 |
| Aether Studios | 186870.0000 | 70.0000 |
| Affray Interactive | 352350.0000 | 82.0000 |
| Afterthought LLC | 2948130.0000 | 79.0000 |

| | | |
|---|---|---|
| Bethesda Game Studios | 3658330 | 74.1667 |
| FromSoftware Inc | 7994620 | 87.8333 |
| CAPCOM Co Ltd | 2728298.571 | 85.2143 |

estimated owners by genre(top left),
estimated owners by developer(top right),
estimated owners for major game companies(bottom)

## Hypothesis 4. Lower prices will increase accessibility, resulting in a larger user base.

- The most expensive games have user bases ranging from approximately 25,000 to 5.7 million.
- The least expensive games are all free-to-play; the user base ranges from 4,000 to over 100 million.
- It is necessary to take into account the massive success achieved by Counter-Strike 2 and PUBG: BATTLEGROUNDS.

```sql
select name, price_usd, estimated_owners
from game_table
order by price_usd desc limit 10;
```

| name | price_usd | estimated_owners |
|---|---|---|
| Indiana Jones and the Great... | 69.99 | 354990 |
| RealFlight Evolution | 69.99 | 24390 |
| CODE VEIN II | 69.99 | 119790 |
| Resident Evil Requiem | 69.99 | 2390010 |
| The Crew Motorfest | 69.99 | 557670 |
| Microsoft Flight Simulator ... | 69.99 | 540120 |
| WWE 2K26 | 69.99 | 25320 |
| Sonic Racing CrossWorlds | 69.99 | 294870 |
| Monster Hunter Stories 3 Tw... | 69.99 | 47100 |
| Monster Hunter Wilds | 69.99 | 5720670 |

```sql
select name, price_usd, estimated_owners
from game_table
order by price_usd asc limit 10;
```

| name | price_usd | estimated_owners |
|---|---|---|
| CounterStrike 2 | 0 | 149410950 |
| Fishing Planet | 0 | 16500 |
| PUBG BATTLEGROUNDS | 0 | 52726470 |
| Neverwinter | 0 | 8880 |
| Horizon Zero Dawn Complete ... | 0 | 2855100 |
| eFootball | 0 | 36270 |
| Apex Legends | 0 | 47310 |
| Call of Duty Warzone | 0 | 293190 |
| Crush Crush | 0 | 16860 |
| Dungeon Defenders II | 0 | 89490 |

Top 10 most expensive games(left) and Top 10 cheapest games(right)

```sql
select name, price_usd, estimated_owners
from game_table
order by estimated_owners desc limit 10;
```

| name | price_usd | estimated_owners |
|---|---|---|
| CounterStrike 2 | 0 | 149410950 |
| PUBG BATTLEGROUNDS | 0 | 52726470 |
| Tom Clancys Rainbow Six Siege | 0 | 37022760 |
| Terraria | 9.99 | 36032190 |
| Rust | 39.99 | 33500670 |
| Garrys Mod | 9.99 | 31657980 |
| Black Myth Wukong | 59.99 | 26084400 |
| Stardew Valley | 14.99 | 25384500 |
| Cyberpunk 2077 | 59.99 | 25070400 |
| The Witcher 3 Wild Hunt | 39.99 | 24271650 |

```sql
select name, price_usd, estimated_owners
from game_table
order by estimated_owners asc limit 10;
```

| name | price_usd | estimated_owners |
|---|---|---|
| Rec Room | 0 | 3300 |
| METAL GEAR SOLID MASTER COL... | 59.99 | 3540 |
| Solateria | 17.99 | 3570 |
| Wireworks | 5.59 | 3990 |
| VRChat | 0 | 4440 |
| AFTERBLAST | 14.99 | 4770 |
| Cargo Hunters | 15.99 | 5040 |
| The Lord of the Rings Online | 0 | 5130 |
| Rewind 99 | 9.59 | 5250 |
| The Liar Princess and the B... | 13.39 | 5280 |

Top 10 with the highest estimated owners(left)
and Top 10 with lowest estimated owners(right)

- In the case of Destiny 2, it is important to consider that the game maintained a certain level of its existing player base after migrating to Steam following its initial success on another platform(Battle.net).
- In the case of “The Sims 4”, it is important to consider that it belongs to the “life simulation” genre and has no direct substitutes.
- When sorted by user count, the PvP FPS genre—rather than free-to-play games—dominates.
- It has been confirmed that the list includes not only low-priced games like ‘Terraria’, ‘Garry’s mod’, and ‘Stardew Valley’, but also high-priced titles such as ‘Rust’, ‘The Witcher 3: Wild Hunt’ and ‘Cyberpunk 2077’.
- Even among games with the smallest user bases, there is a diverse range of pricing, spanning from free to high-priced titles.
- It appears there is no direct correlation between a game’s price and its commercial success.

### Hypothesis 5. Games released longer ago will have higher exposure, leading to a larger user base.

- Many of the games were released within the last year or two (2025 - 2026).
- It is also evident that some titles released in the 2010s, such as ‘VRChat’ and ‘The Lord of the Rings Online’, are included.
- This indicates that there is no direct correlation between a game’s release date and its player count.

```
select name, release_date, estimated_owners
from game_table
order by estimated_owners asc limit 10;
```

| name | release_date | estimated_owners |
|---|---|---|
| Rec Room | 2021-09-02 | 3300 |
| METAL GEAR SOLID MASTER COL... | 2023-10-24 | 3540 |
| Solateria | 2026-03-12 | 3570 |
| Wireworks | 2026-03-09 | 3990 |
| VRChat | 2017-02-01 | 4440 |
| AFTERBLAST | 2025-11-26 | 4770 |
| Cargo Hunters | 2026-03-09 | 5040 |
| The Lord of the Rings Online | 2012-06-06 | 5130 |
| Rewind 99 | 2026-03-11 | 5250 |
| The Liar Princess and the B... | 2026-03-11 | 5280 |

Release dates of the 10 games with the fewest players

## Project Retrospective and Areas for Improvement

1. **DataFrame Selection**

- Failed to identify issues **inherent in the CSV** file beforehand.
- Realized the necessity of conducting a thorough **preliminary check when selecting the DataFrame.**
- Recognized the need for **data cleaning techniques** to handle situations where the DataFrame cannot be modified.

2. **Lack of proficiency in grammar and analytical methods during the analysis**

- Since a single game can span multiple genres, **duplicate game titles occurred.**
- Realized too late that I should have implemented **grouping via 'GROUP BY'** or **prevented duplicates using 'DISTINCT'**.

# Data Engineering

## Animal Shelter Monitoring system (Team Project)

**GitHub:** https://github.com/yehyun-42/yehyun-42.github.io/tree/main/Project/Team_Project

### Tech Stack

| | |
|---|---|
| Language | Python |
| IDE | VS Code |
| Database | SQLAlchemy, Flask-Migrate |
| Web streaming server | Flask |

**Problem Identification**

- While volunteering at a **private shelter for abandoned dogs**, experienced firsthand the staffing shortages faced by a very small team managing a large number of animals.
- Began seeking technological solutions to address the various problems caused by this lack of manpower.

**Technical Motivation**

- Observed **deep learning-based object detection technology currently in practical use** at the Gwacheon Korea National Science Museum.
- Aimed to build a system that alleviates human workload by applying the technology learned in the classroom to real-world scenarios.

**Reasons for Selection the Topic**

- Establishing protocols to handle unexpected incidents and crises arising from staff shortages in animal shelters, where a small team manages a large number of animals.
- Aiming for more efficient management and reduced human workload by utilizing drones and AI.

**Benchmarking Reference**

- **Technology for Estimating Penguin Populations by the Korea Sejong Station Wintering Research Team.**
  A technology that uses drones to capture footage of penguin habitats and employs AI to identify penguin populations and nests. This serves as a reference for developing capabilities to detect the populations of protected animal species.

- **Smart roadkill prevention systems installed in areas such as Korea Yangpyeong-gun, Gyeonggi-do, and Pyeongchang-gun, Gangwon-do.**
  A system designed to prevent wildlife roadkill on national highways. It utilizes AI-based CCTV and radar sensors to detect the presence of wildlife. This information serves as a reference for implementing wildlife detection capabilities.
- **Real-time mobile push notifications from the home camera.**
  A home camera function that monitors a designated area and sends an alert to the user's smartphone (or similar device) when specific conditions are met; useful for reference when setting up a real-time notification system.

**Role in Project: Backend and DB pipeline development**

- ***Database & Data Management***
  *Database creation and visualization*
  *Integration and Management of Detection Log Database*
- Streaming Troubleshooting Fix
  Resolving ESP32-CAM Streaming Errors
- Alert Notification Function
  E-mail Alerts: E-mail integration
  Discord Alerts: Webhook integration

**Data Flow**

Threshold Web Dashboard → Detection Threshold DB → Threshold Dictionary → YOLO (Threshold-based Detection) → Detection History DB → Detection History Web Dashboard

# Retrospective

## Reflection and Improvement

1. **Population Detection System**
   Shelters typically house dozens to hundreds of animals.
   **Current detection logic**: Data is saved to the database and an alarm is triggered only if the "below-threshold" state persists for 10 seconds after initial detection.
   This approach is difficult to implement in actual shelter environments; a more reliable method of object recognition is required.
   Techniques such as partial labeling could potentially resolve issues related to object overlapping.
2. **Wild Animal Detection System**
   **Current detection logic**: Detects only previously registered wildlife.
   Wildlife may change habitats for various reasons.
   The appearance of new wildlife in the area entails the inconvenience of requiring a learning and registration process.

Logic of the benchmarked smart roadkill prevention system: Determines whether a detected moving object is wildlife.
The logic needs to be modified to detect “wildlife” itself, rather than just “registered wildlife”, by utilizing this approach.

3. **Map Service**
   **Current detection logic**: An alarm is triggered when wildlife appear.
   To devise a fundamental solution, a map recording Wildlife sighting locations is provided.
   It is necessary to identify key wildlife hotspots and establish a means to share information on how to deal with wildlife.

## Growth

**Experience in database development and integration**

- Designed a data pipeline connecting the ESP32-CAM, Flask server, YOLO, and the database.
- Designed and implemented database tables tailored to specific requirements.
- Gained experience in the server-side process of storing and visualizing YOLO detection data, as well as the process of storing user-inputted information and linking it with reference thresholds.

**Troubleshooting and Prevention**

- Encountered and resolved unexpected software-related issues.
- Prevented potential future errors and enhanced security.

**Communication between team members**

- Instead of grappling with project-related issues in isolation, determine the direction and resolve them through communication with the entire team.

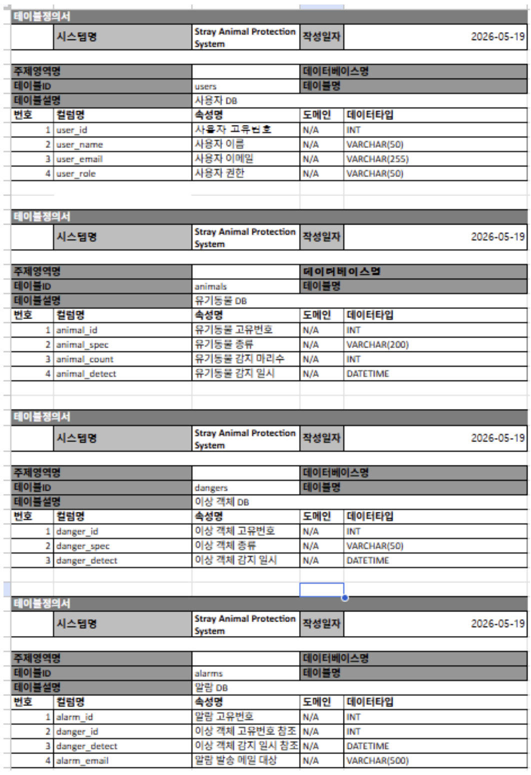
## Database & Data Management

### Initial Concept Table

| | |
|---|---|
| User | Users |
| Detected History of Protected Species | Animals |
| Detection History of Wild animal | Dangers |
| Alert Sent | Alarms |

### Significant

- For the alarm generation database, it is necessary to reference records from other databases.
- Reference the unique ID and detection timestamp from the wild animal detection database(Danger DB).
- Reference the email address from the User database.



| 테이블정의서 | | | | |
|---|---|---|---|---|
| | 시스템명 | Stray Animal Protection System | 작성일자 | 2026-05-19 |
| 주제영역명 | | | 데이터베이스명 | |
| 테이블ID | | users | 테이블명 | |
| 테이블설명 | | 사용자 DB | | |
| 번호 | 컬럼명 | 속성명 | 도메인 | 데이터타입 |
| 1 | user_id | 사용자 고유번호 | N/A | INT |
| 2 | user_name | 사용자 이름 | N/A | VARCHAR(50) |
| 3 | user_email | 사용자 이메일 | N/A | VARCHAR(255) |
| 4 | user_role | 사용자 권한 | N/A | VARCHAR(50) |

| 테이블정의서 | | | | |
|---|---|---|---|---|
| | 시스템명 | Stray Animal Protection System | 작성일자 | 2026-05-19 |
| 주제영역명 | | | 데이터베이스명 | |
| 테이블ID | | animals | 테이블명 | |
| 테이블설명 | | 유기동물 DB | | |
| 번호 | 컬럼명 | 속성명 | 도메인 | 데이터타입 |
| 1 | animal_id | 유기동물 고유번호 | N/A | INT |
| 2 | animal_spec | 유기동물 종류 | N/A | VARCHAR(200) |
| 3 | animal_count | 유기동물 감지 마리수 | N/A | INT |
| 4 | animal_detect | 유기동물 감지 일시 | N/A | DATETIME |

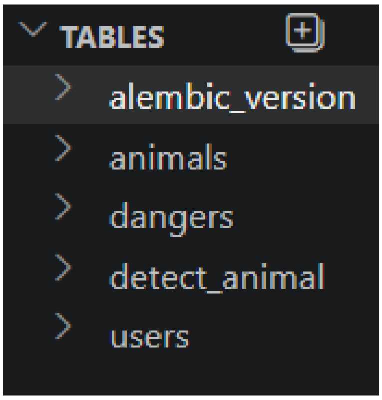
| 테이블정의서 | | | | |
|---|---|---|---|---|
| | 시스템명 | Stray Animal Protection System | 작성일자 | 2026-05-19 |
| 주제영역명 | | | 데이터베이스명 | |
| 테이블ID | | dangers | 테이블명 | |
| 테이블설명 | | 이상 객체 DB | | |
| 번호 | 컬럼명 | 속성명 | 도메인 | 데이터타입 |
| 1 | danger_id | 이상 객체 고유번호 | N/A | INT |
| 2 | danger_spec | 이상 객체 종류 | N/A | VARCHAR(50) |
| 3 | danger_detect | 이상 객체 감지 일시 | N/A | DATETIME |

| 테이블정의서 | | | | |
|---|---|---|---|---|
| | 시스템명 | Stray Animal Protection System | 작성일자 | 2026-05-19 |
| 주제영역명 | | | 데이터베이스명 | |
| 테이블ID | | alarms | 테이블명 | |
| 테이블설명 | | 알람 DB | | |
| 번호 | 컬럼명 | 속성명 | 도메인 | 데이터타입 |
| 1 | alarm_id | 알람 고유번호 | N/A | INT |
| 2 | danger_id | 이상 객체 고유번호 참조 | N/A | INT |
| 3 | danger_detect | 이상 객체 감지 일시 참조 | N/A | DATETIME |
| 4 | alarm_email | 알람 발송 메일 대상 | N/A | VARCHAR(500) |

Table Definition Document

### Final confirmed Table

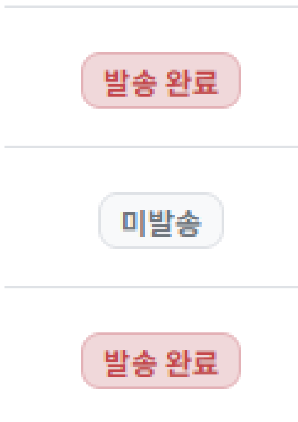
| | |
|---|---|
| User | Users |
| Protected Species count | Animals |
| Wild animal that need Detection | Dangers |
| Detection History | Detect_animal |




Final confirmed DB

### Reason for table modification

- Identified the need to track the population count(baseline) of sheltered stray animals.
- To improve efficiency, modified the system to evaluate detection data in real-time and determine whether to trigger an alert, rather than relying on a separate, tactically managed alert dispatch table.
- Configure the system to send an alert when specific conditions (e.g., population falling below the threshold) are met.




Visualization of boolean values

Added a 'detect_alarm' column with a boolean value to visually indicate whether an alert has been triggered.
Enabled status monitoring—specifically "Not Sent" vs "Sent"—within the visualization dashboard.

### Preventing Circular Reference

Initial DB configuration structure: importing the database within the 'app.py' file.
Identified potential risks of circular dependencies and maintenance issues as the monitoring system scales.

**Solution**: Create an extensions.py file and set it to be responsible only for the DB. Afterwards, import extensions.py rather than DB in app.py.

extensions.py
project_SSA > apps > extensions.py > ...

```python
from flask_sqlalchemy import SQLAlchemy

db=SQLAlchemy()
```

extensions.py for Preventing Circular Reference

### Database Visualization Dashboard

Visualize the database to facilitate easy access and management.
Visualize the stored data by applying filtering and processing, enabling users to intuitively understand the information.

The 'Animals' table is integrated to allow for immediate updates within the visualization dashboard.

- The protected animal (key) and its population count (value) are defined as a dictionary.
- The system is configured so that user inputs are applied to the dictionary.

### Issues encountered during table integration

- Unable to link YOLO detection records and the 'Detect_animal' table with the threshold dictionary and the 'Animals' table.
  **Cause**: Tensor values and dictionary values were not converted into a format compatible with the database table.
  **Solution**: Directly extract the Tensor values and dictionary column values, and insert them directly into SQLite.